$c = \frac{d'}{t}$; rewritten for t as, $t = \frac{d'}{c}$

As the time taken in both the cases is the same, we can equate the time t,

$$\frac{d'}{c} = \frac{d}{v}$$

Rewritten for the optical path d' as, $d' = \frac{c}{v}d$

As, $\frac{c}{v} = n$; the optical path d' is,

$$d' = nd \qquad (6.17)$$

The value of n is always greater than 1, for a medium. Thus, the optical path d' of a medium is always greater than d.

EXAMPLE 6.6

Light travels from air into a glass slab of thickness 50 cm and refractive index 1.5.

(a) What is the speed of light in the glass slab?

(b) What is the time taken by the light to travel through the glass slab?

(c) What is the optical path of the glass slab?

Solution

Given, thickness of glass slab, d = 50 cm = 0.5 m, refractive index, n = 1.5

refractive index, $n = \frac{c}{v}$

(a) speed of light in the glass slab is,

$$v = \frac{c}{n} = \frac{3\times10^8}{1.5} = 2\times10^8\,\text{m s}^{-1}$$

(b) time taken by light to travel through the glass slab is,

$$t = \frac{d}{v} = \frac{0.5}{2\times10^8} = 2.5\times10^{-9}\,\text{s}$$

(c) optical path,

$$d' = nd = 1.5\times0.5 = 0.75\,\text{m} = 75\,\text{cm}$$

Light would have travelled an additional 25 cm (75 cm – 50 cm) in vacuum at the same time had there been no glass slab in its path.

6.4 REFRACTION

Refraction is passing of light from one optical medium to another optical medium through a boundary. In refraction, the angle of incidence i in one medium and the angle of refraction r in the other medium are measured with respect to the normal drawn to the surface at the point of incidence of light. According to laws of refraction,

(i) The incident ray, refracted ray and normal to the refracting surface are all coplanar (ie. lie in the same plane).

(ii) The ratio of sine of angle of incident i in the first medium to the sine of angle of refraction r in the second medium is equal to the ratio of refractive index n_2 of the second medium to the refractive index n_1 of the first medium.

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} \qquad (6.18)$$

The above equation is in the ratio form. It can also be written in a product form as,

$$n_1 \sin i = n_2 \sin r \qquad (6.19)$$

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